Data Storing

To begin analysis for my applied project. I had to create a bucket in s3, and then create database in athena using the following code:

```
CREATE DATABASE airlinedelay;
CREATE EXTERNAL TABLE airlinedelay (
    year INT,
    month INT,
    carrier STRING,
    carrier_name STRING,
    airport STRING,
    airport_name STRING,
    arr_flights FLOAT,
    arr_del15 FLOAT,
    carrier_ct FLOAT,
    weather_ct FLOAT,
    nas_ct FLOAT,
    security_ct FLOAT,
    late_aircraft_ct FLOAT,
    arr_cancelled FLOAT,
    arr_diverted FLOAT,
    arr_delay FLOAT,
    carrier_delay FLOAT,
    weather_delay FLOAT,
    nas_delay FLOAT,
    security_delay FLOAT,
    late_aircraft_delay FLOAT
)
ROW FORMAT SERDE 'org.apache.hadoop.hive.serde2.lazy.LazySimpleSerDe'
WITH SERDEPROPERTIES (
    'serialization.format' = ',',
    'field.delim' = ','
)
LOCATION 's3://chisanshiairlinedelay/'
TBLPROPERTIES ('has_encrypted_data'='false');
```

Exploratory Data Analysis

To get the structure of the table and the data types involved:

```
DESCRIBE airlinedelay;
```

```
year                    int
month                   int
carrier                 string
carrier_name            string
airport                 string
airport_name            string
arr_flights             float
arr_del15               float
carrier_ct              float
weather_ct              float
nas_ct                  float
security_ct             float
late_aircraft_ct        float
arr_cancelled           float
arr_diverted            float
arr_delay               float
carrier_delay           float
weather_delay           float
nas_delay               float
security_delay          float
late_aircraft_delay     float
```

Columns:
Year- Year data collected

Month- Numeric representation of the month

Carrier- Carrier.

Carrier_name- Carrier Name.

Airport- Airport code.

Airport_name- Name of airport.

Arr_flights- Number of flights arriving at airport

Arr_del15- Number of flights more than 15 minutes late

Carrier_ct- Number of flights delayed due to air carrier. (e.g. no crew)

Weather_ct- Number of flights due to weather.

Nas_ct- Number of flights delayed due to National Aviation System (e.g. heavy air traffic).

Security_ct- Number of flights canceled due to a security breach.

Late_aircraft_ct- Number of flights delayed as a result of another flight on the same aircraft delayed

Arr_cancelled- Number of cancelled flights

Arr_diverted- Number of flights that were diverted

Arr_delay- Total time (minutes) of delayed flight.

Carrier_delay- Total time (minutes) of delay due to air carrier

Weather_delay- Total time (minutes) of delay due to inclement weather.

Nas_delay- Total time (minutes) of delay due to National Aviation System.

Security_delay- Total time (minutes) of delay as a result of a security issue .

Late_aircraft_delay- Total time (minutes) of delay flights as a result of a previous flight on the same airplane being late.

Missing Values

Checking for missing values in key columns, I only focused on the columns that had to do with delayals because that was the key focus of my project, these included Arr_delay, Carrier_delay,Weather_delay,Nas_delay and Security_delay:

```
SELECT
    SUM(CASE WHEN arr_delay IS NULL THEN 1 ELSE 0 END) AS
missing_values,
    COUNT(*) AS total_records,
    (SUM(CASE WHEN arr_delay IS NULL THEN 1 ELSE 0 END) * 100.0
/ COUNT(*)) AS missing_percentage
FROM airlinedelay;
```

Search rows

#	missing_values	total_records	missing_percentage
1	9	3352	0.2684964200477327

```
SELECT
    SUM(CASE WHEN carrier_delay IS NULL THEN 1 ELSE 0 END) AS
missing_values,
    COUNT(*) AS total_records,
```

```
    (SUM(CASE WHEN carrier_delay IS NULL THEN 1 ELSE 0 END) * 100.0 /
COUNT(*)) AS missing_percentage
FROM airlinedelay;
```

#	missing_values	total_records	missing_percentage
1	9	3352	0.2684964200477327

```
SELECT
    SUM(CASE WHEN weather_delay IS NULL THEN 1 ELSE 0 END) AS
missing_values,
    COUNT(*) AS total_records,
    (SUM(CASE WHEN weather_delay IS NULL THEN 1 ELSE 0 END) * 100.0 /
COUNT(*)) AS missing_percentage
FROM airlinedelay;
```

#	missing_values	total_records	missing_percentage
1	9	3352	0.2684964200477327

```
SELECT
    SUM(CASE WHEN nas_delay IS NULL THEN 1 ELSE 0 END) AS
missing_values,
    COUNT(*) AS total_records,
    (SUM(CASE WHEN nas_delay IS NULL THEN 1 ELSE 0 END) * 100.0 /
COUNT(*)) AS missing_percentage
FROM airlinedelay;
```

#	missing_values	total_records	missing_percentage
1	9	3352	0.2684964200477327

```
SELECT
    SUM(CASE WHEN security_delay IS NULL THEN 1 ELSE 0 END) AS
missing_values,
    COUNT(*) AS total_records,
    (SUM(CASE WHEN security_delay IS NULL THEN 1 ELSE 0 END) * 100.0
/ COUNT(*)) AS missing_percentage
FROM airlinedelay;
```

#	missing_values	total_records	missing_percentage
1	9	3352	0.2684964200477327

```
SELECT
    SUM(CASE WHEN late_aircraft_delay IS NULL THEN 1 ELSE 0 END) AS
missing_values,
    COUNT(*) AS total_records,
    (SUM(CASE WHEN late_aircraft_delay IS NULL THEN 1 ELSE 0 END) *
100.0 / COUNT(*)) AS missing_percentage
FROM airlinedelay;
```

Search rows

#	missing_values	total_records	missing_percentage
1	9	3352	0.2684964200477327

I did not drop data because it wasn't that big to affect overall analysis of data, the missing percentage is very low ,<5%, dropping rows may not significantly impact analysis.
**Query calculations are off, but the missing percentage is about 0.27%*

Analysis

1. **Airline Performance**: Which airlines experience the most frequent and longest delays?

   Top 10 Airlines that experience the most frequent delays

```
SELECT

    carrier_name,

    COUNT(*) as total_delays

FROM

    airlinedelay

WHERE

    arr_del15 > 0

GROUP BY

    carrier_name

ORDER BY

    total_delays DESC
```

```
LIMIT 10;
```

#	carrier_name	total_delays
1	SkyWest Airlines Inc.	479
2	Envoy Air	285
3	Delta Air Lines Inc.	252
4	Allegiant Air	240
5	Endeavor Air Inc.	226
6	Mesa Airlines Inc.	219
7	American Airlines Inc.	199
8	Republic Airline	196
9	United Air Lines Inc.	191
10	PSA Airlines Inc.	187

Skywest Airlines Inc experienced the most frequent delays, It had 479 flights which were more than 15 minutes late.Envoy Air and Delta Air Lines Inc. followed with 285 and 252 delays, respectively. The variation in total delays suggests differences in operational efficiency, flight volume, or external factors affecting airline delays.

Top 10 longest average delays by airline

```
SELECT
    carrier_name,
    AVG(arr_delay) as avg_delay
FROM
    airlinedelay
WHERE
    arr_delay > 0
```

```
GROUP BY
    carrier_name
ORDER BY
    avg_delay DESC
    LIMIT 10;
```

#	carrier_name	avg_delay
1	American Airlines Inc.	3.2586207
2	SkyWest Airlines Inc.	2.764045
3	Southwest Airlines Co.	2.7567568
4	Mesa Airlines Inc.	2.6060605
5	Alaska Airlines Inc.	2.5
6	Republic Airline	2.5
7	Envoy Air	2.36
8	United Air Lines Inc.	2.2692308
9	JetBlue Airways	2.25
10	PSA Airlines Inc.	2.1190476

American Airlines Inc. had the highest average delay of approximately 3.26 minutes. SkyWest Airlines Inc. and Southwest Airlines Co. follow closely with average delays of 2.76 and 2.76 minutes, respectively. While SkyWest had the highest total delays in the previous query, American Airlines leads in average delay duration, suggesting that its delays might be more prolonged per occurrence.

2. **Time-based Trends**: How do delays vary by season, month, day of the week, and time of day?

```
SELECT
    month,
```

```
    COUNT(*) as total_delays,
    AVG(arr_delay) as avg_delay
FROM
    airlinedelay
WHERE
    arr_del15 > 0
GROUP BY
    month
ORDER BY
    Month;
```

#	month	total_delays	avg_delay
1	12	3343	0.5758301

The dataset only contains data from December, it is not easy to tell how the delayals vary by season or month

3. **Airport Impact**: Which airports have the highest average delay times, and are certain routes more prone to delays?

Top 10 Airports with highest average delay times

```
SELECT
    airport_name,
    AVG(arr_delay) as avg_delay
FROM
    airlinedelay
WHERE
    arr_delay > 0
```

```
GROUP BY
    airport_name
ORDER BY
    avg_delay DESC
    LIMIT 10;
```

#	airport_name	avg_delay
1	"Aspen	31.5
2	"Sun Valley/Hailey/Ketchum	23.0
3	"Dallas/Fort Worth	9.294118
4	"Dallas	9.0
5	"San Francisco	7.125
6	"Juneau	7.0
7	"Baltimore	6.6666665
8	"Long Beach	6.0
9	"Houston	5.923077
10	"Salt Lake City	5.857143

Aspen experienced the highest average delay of 31.5 minutes, followed by SunValley/Hailey/Ketchum at 23.0 minutes. Dallas/Fort Worth and San Francisco have much lower average delays, around 9.29 and 7.13 minutes, respectively. The relatively lower delays at airports like Houston (5.92 minutes) and Salt Lake City (5.86 minutes) suggest better operational efficiency or fewer external disruptions

Routes more prone to delays

The dataset does not contain origin and arrival destination.

4. **Delay Causes**: What are the most common causes of delays, and how do they differ by airline and airport?

    Causes of Delay General

```
SELECT
    'carrier_delay' as cause,
    SUM(carrier_delay) as total_delay_time
FROM
    airlinedelay
UNION ALL
SELECT
    'weather_delay',
    SUM(weather_delay)
FROM
    airlinedelay
UNION ALL
SELECT
    'nas_delay',
    SUM(nas_delay)
FROM
    airlinedelay
UNION ALL
SELECT
    'security_delay',
    SUM(security_delay)
FROM
    airlinedelay
UNION ALL
SELECT
    'late_aircraft_delay',
```

```
        SUM(late_aircraft_delay)
    FROM
        airlinedelay
    ORDER BY
        total_delay_time DESC;
```

#	cause	total_delay_time
1	carrier_delay	1.1145121E7
2	weather_delay	3826943.0
3	security_delay	2505844.0
4	nas_delay	593688.0
5	late_aircraft_delay	18055.0

Carrier-related delays are the most significant cause of airline delay, with a total delay time exceeding 11.1 million minutes. Weather delays follow as the second largest cause, accounting for approximately 3.8 million minutes, which suggests that environmental factors play a major role in flight disruptions. Security and NAS (National Aviation System) delays contribute significantly as well, with 2.5 million and 593,688 minutes, respectively, indicating potential inefficiencies in airport security procedures and air traffic management. Late aircraft delays, which result from previous flight delays affecting subsequent flights, are the least impactful in comparison, with only 18,055 minutes in total.

Delay Causes per Airline (Explanation of results in visualization)

```
SELECT
    carrier_name,
    'carrier_delay' as cause,
    SUM(carrier_delay) as total_delay_time
FROM
    airlinedelay
```

```
GROUP BY
    carrier_name
UNION ALL
SELECT
    carrier_name,
    'weather_delay',
    SUM(weather_delay)
FROM
    airlinedelay
GROUP BY
    carrier_name
UNION ALL
SELECT
    carrier_name,
    'nas_delay',
    SUM(nas_delay)
FROM
    airlinedelay
GROUP BY
    carrier_name
UNION ALL
SELECT
    carrier_name,
    'security_delay',
    SUM(security_delay)
FROM
```

```sql
    airlinedelay
GROUP BY
    carrier_name
UNION ALL
SELECT
    carrier_name,
    'late_aircraft_delay',
    SUM(late_aircraft_delay)
FROM
    airlinedelay
GROUP BY
    carrier_name
ORDER BY
    carrier_name,
    total_delay_time DESC;
```

#	carrier_name	cause	total_delay_time
1	Alaska Airlines Inc.	carrier_delay	351035.0
2	Alaska Airlines Inc.	security_delay	107680.0
3	Alaska Airlines Inc.	weather_delay	100526.0
4	Alaska Airlines Inc.	nas_delay	7656.0
5	Alaska Airlines Inc.	late_aircraft_delay	1787.0
6	Allegiant Air	carrier_delay	240714.0
7	Allegiant Air	weather_delay	79398.0
8	Allegiant Air	security_delay	48560.0
9	Allegiant Air	nas_delay	16220.0
10	Allegiant Air	late_aircraft_delay	883.0
11	American Airlines Inc.	carrier_delay	1185345.0
12	American Airlines Inc.	weather_delay	448469.0
13	American Airlines Inc.	security_delay	271857.0
14	American Airlines Inc.	nas_delay	39269.0
15	American Airlines Inc.	late_aircraft_delay	1921.0
16	Delta Air Lines Inc.	carrier_delay	1077429.0
17	Delta Air Lines Inc.	weather_delay	423997.0
18	Delta Air Lines Inc.	security_delay	319153.0
19	Delta Air Lines Inc.	nas_delay	62238.0
20	Delta Air Lines Inc.	late_aircraft_delay	1150.0
21	Endeavor Air Inc.	carrier_delay	431683.0
22	Endeavor Air Inc.	weather_delay	142835.0
23	Endeavor Air Inc.	security_delay	98934.0
24	Endeavor Air Inc.	nas_delay	32914.0
25	Endeavor Air Inc.	late_aircraft_delay	248.0
26	Envoy Air	carrier_delay	505165.0
27	Envoy Air	security_delay	125225.0
28	Envoy Air	weather_delay	111682.0
29	Envoy Air	nas_delay	53187.0
30	Envoy Air	late_aircraft_delay	646.0
31	ExpressJet Airlines LLC	carrier_delay	254148.0
32	ExpressJet Airlines LLC	security_delay	87490.0
33	ExpressJet Airlines LLC	weather_delay	79871.0
34	ExpressJet Airlines LLC	nas_delay	8325.0
35	ExpressJet Airlines LLC	late_aircraft_delay	0.0
36	Frontier Airlines Inc.	carrier_delay	232696.0
37	Frontier Airlines Inc.	weather_delay	70578.0
38	Frontier Airlines Inc.	security_delay	59817.0

#	carrier_name	cause	total_delay_time
39	Frontier Airlines Inc.	nas_delay	2744.0
40	Frontier Airlines Inc.	late_aircraft_delay	0.0
41	Hawaiian Airlines Inc.	carrier_delay	39243.0
42	Hawaiian Airlines Inc.	weather_delay	29003.0
43	Hawaiian Airlines Inc.	nas_delay	1956.0
44	Hawaiian Airlines Inc.	security_delay	412.0
45	Hawaiian Airlines Inc.	late_aircraft_delay	159.0
46	JetBlue Airways	carrier_delay	864790.0
47	JetBlue Airways	weather_delay	297439.0
48	JetBlue Airways	security_delay	192786.0
49	JetBlue Airways	nas_delay	20295.0
50	JetBlue Airways	late_aircraft_delay	847.0
51	Mesa Airlines Inc.	carrier_delay	530595.0
52	Mesa Airlines Inc.	weather_delay	184419.0
53	Mesa Airlines Inc.	security_delay	67264.0
54	Mesa Airlines Inc.	nas_delay	24587.0
55	Mesa Airlines Inc.	late_aircraft_delay	596.0
56	PSA Airlines Inc.	carrier_delay	479952.0
57	PSA Airlines Inc.	weather_delay	137901.0
58	PSA Airlines Inc.	security_delay	73460.0
59	PSA Airlines Inc.	nas_delay	18088.0
60	PSA Airlines Inc.	late_aircraft_delay	787.0
61	Republic Airline	carrier_delay	541222.0
62	Republic Airline	security_delay	177608.0
63	Republic Airline	weather_delay	130539.0
64	Republic Airline	nas_delay	23656.0
65	Republic Airline	late_aircraft_delay	705.0
66	SkyWest Airlines Inc.	carrier_delay	1804000.0
67	SkyWest Airlines Inc.	weather_delay	788943.0
68	SkyWest Airlines Inc.	security_delay	219743.0
69	SkyWest Airlines Inc.	nas_delay	219602.0
70	SkyWest Airlines Inc.	late_aircraft_delay	1747.0
71	Southwest Airlines Co.	carrier_delay	1396083.0
72	Southwest Airlines Co.	weather_delay	506268.0
73	Southwest Airlines Co.	security_delay	224360.0

74	Southwest Airlines Co.	nas_delay	15075.0
75	Southwest Airlines Co.	late_aircraft_delay	4694.0
76	Spirit Air Lines	carrier_delay	307232.0
77	Spirit Air Lines	security_delay	136959.0
78	Spirit Air Lines	weather_delay	73385.0
79	Spirit Air Lines	nas_delay	12885.0
80	Spirit Air Lines	late_aircraft_delay	1767.0
81	United Air Lines Inc.	carrier_delay	903789.0
82	United Air Lines Inc.	security_delay	294536.0
83	United Air Lines Inc.	weather_delay	221690.0
84	United Air Lines Inc.	nas_delay	34991.0
85	United Air Lines Inc.	late_aircraft_delay	118.0
86	carrier_name	security_delay	

5. **Weather Influence**: How does weather impact flight delays, and can it be correlated with delay frequency?

```
SELECT

    weather_ct,

    COUNT(*) as total_flights,

    AVG(arr_delay) as avg_delay

FROM

    airlinedelay

GROUP BY

    weather_ct

ORDER BY

    Weather_ct;
```

#	weather_ct	total_flights	avg_delay
1	0.0	311	0.057877813
2	0.01	2	0.0
3	0.02	2	0.0
4	0.03	3	1.0
5	0.04	5	0.0
6	0.05	2	0.0
7	0.06	8	0.0
8	0.08	5	0.0
9	0.09	3	0.0
10	0.1	2	0.0
90	0.98	7	0.5714286
91	0.99	3	2.0
92	1.0	167	0.1257485
93	1.01	1	0.0
94	1.03	6	0.5
95	1.04	3	0.33333334
96	1.05	1	0.0
97	1.06	3	1.0
98	1.07	2	0.0
99	1.08	2	0.0
100	1.09	2	0.0

I got the first 10 and last 10 of the output of the query. Weather significantly impacts flight delays, but its correlation with delay frequency depends on specific conditions. When weather-related delay counts (weather_ct) are low, the average delay is also low, likely due to minimal weather disruptions. In some cases, a lower weather_ct corresponds to fewer total flights, but this trend is not consistent across all instances. This suggests that while weather is a contributing factor to delays, it is not the sole determinant. weather disruptions can increase the frequency and duration of delays, but their impact varies.

6. **Forecasting Trends**: Is there a trend in delay duration over time, and can we forecast future delays?

```
SELECT
    year,
    month,
```

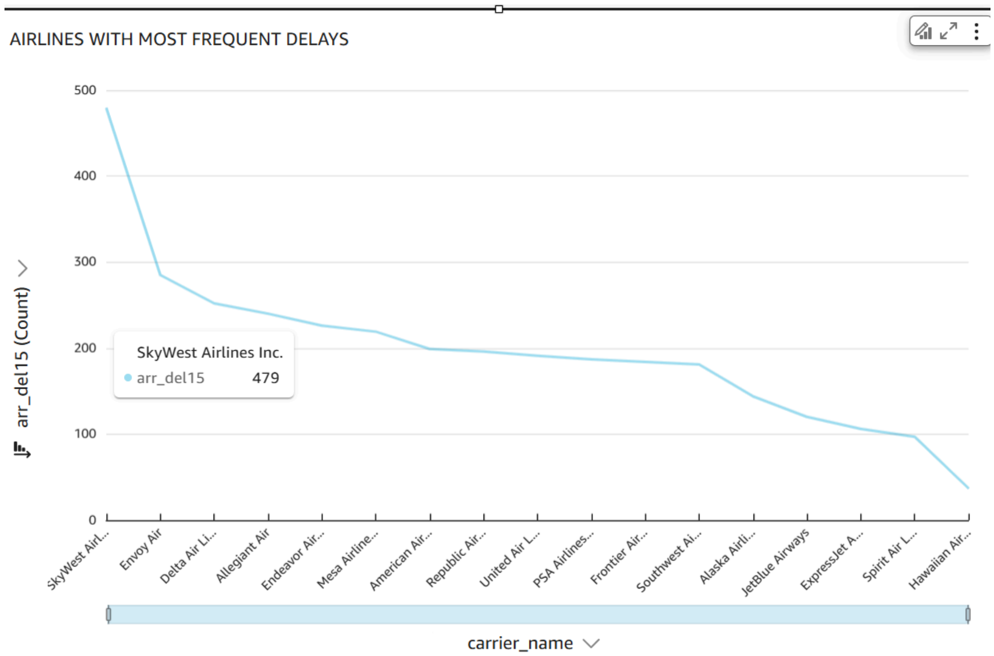
```
    AVG(arr_delay) as avg_delay
FROM
    airlinedelay
WHERE
    arr_delay > 0
GROUP BY
    year,
    month
ORDER BY
    year,
    month;
```

#	year	month	avg_delay
1	2019	12	2.7051792
2	2020	12	1.8590164

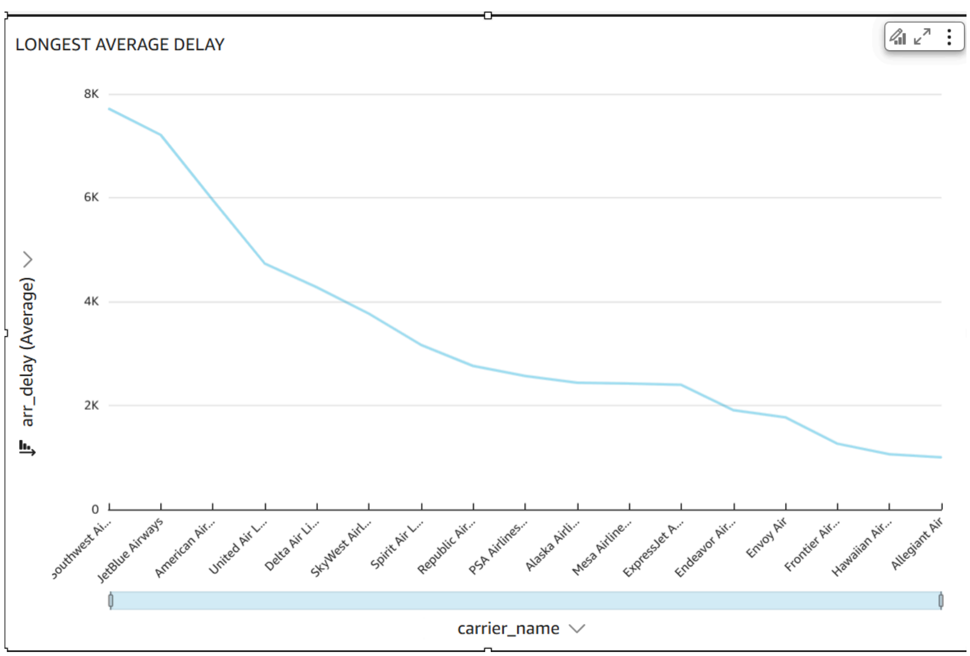
The dataset only contains flight information from December of 2019 and 2020. There is a decrease in delays from 2.71 minutes in 2019 to 1.86 minutes in 2020. This reduction could be attributed to the impact of the COVID-19 pandemic, which significantly reduced air traffic in 2020, leading to fewer delays due to lower congestion in airports and airspace.

QUICKSIGHT VISUALIZATIONS

1. **Airline Performance**: Which airlines experience the most frequent and longest delays?

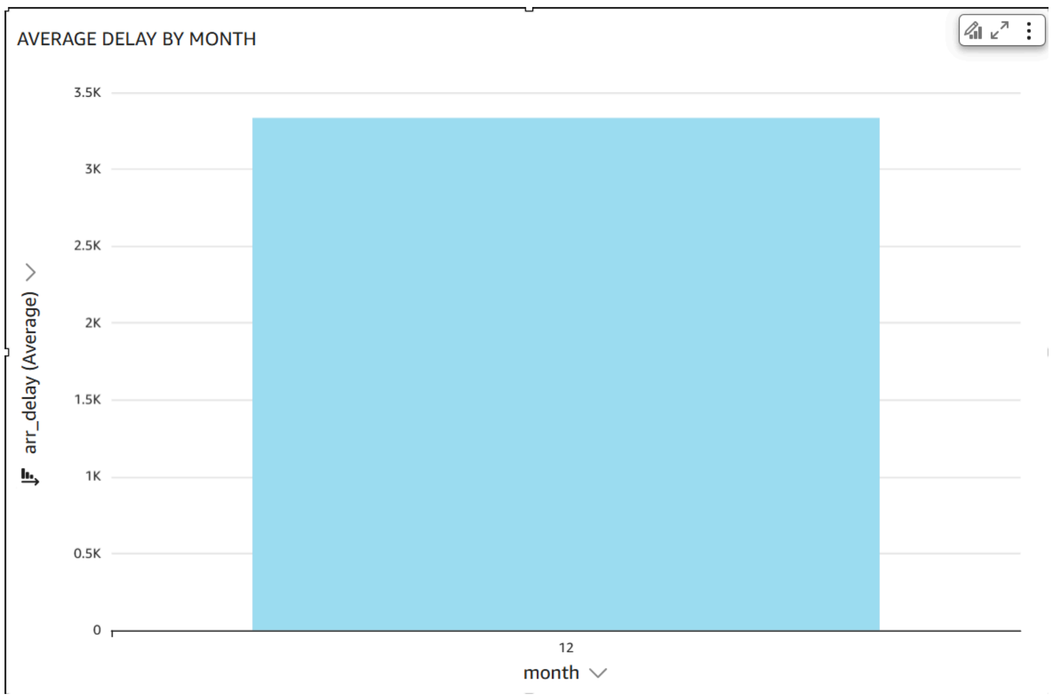


Most frequent delays: Skywest Airlines Inc experienced the most frequent delays, It had 479 flights which were more than 15 minutes late.Envoy Air and Delta Air Lines Inc. followed with 285 and 252 delays, respectively. The variation in total delays suggests differences in operational efficiency, flight volume, or external factors affecting airline delays.



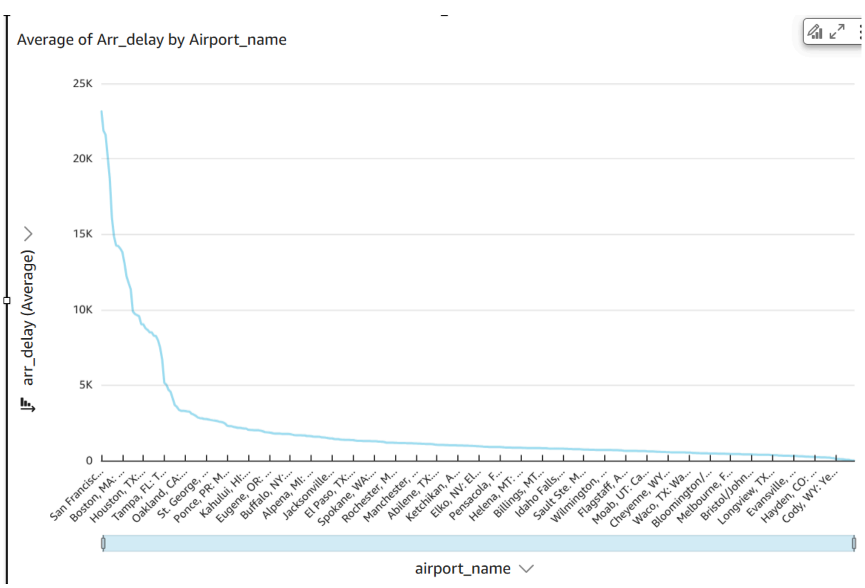
Longest Average Delay: Southwest Airlines experienced the longest delays, followed by JetBlue Airways, American Airlines, and United Airlines. The trend shows a gradual decline in average delay times as we move across different airlines,

2. **Time-based Trends**: How do delays vary by season, month, day of the week, and time of day?



The data set only contained data from the month of December, so it is not possible to see the variance in delays by season and month.

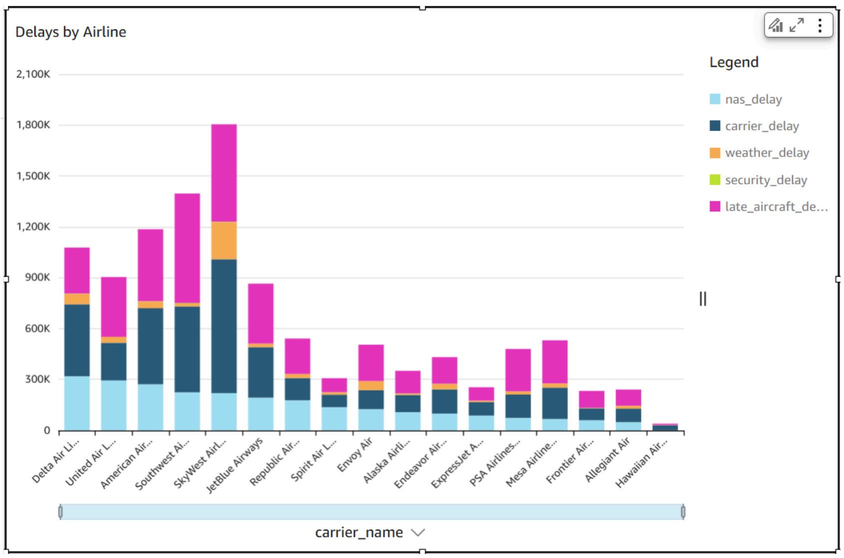
3. **Airport Impact**: Which airports have the highest average delay times, and are certain routes more prone to delays?



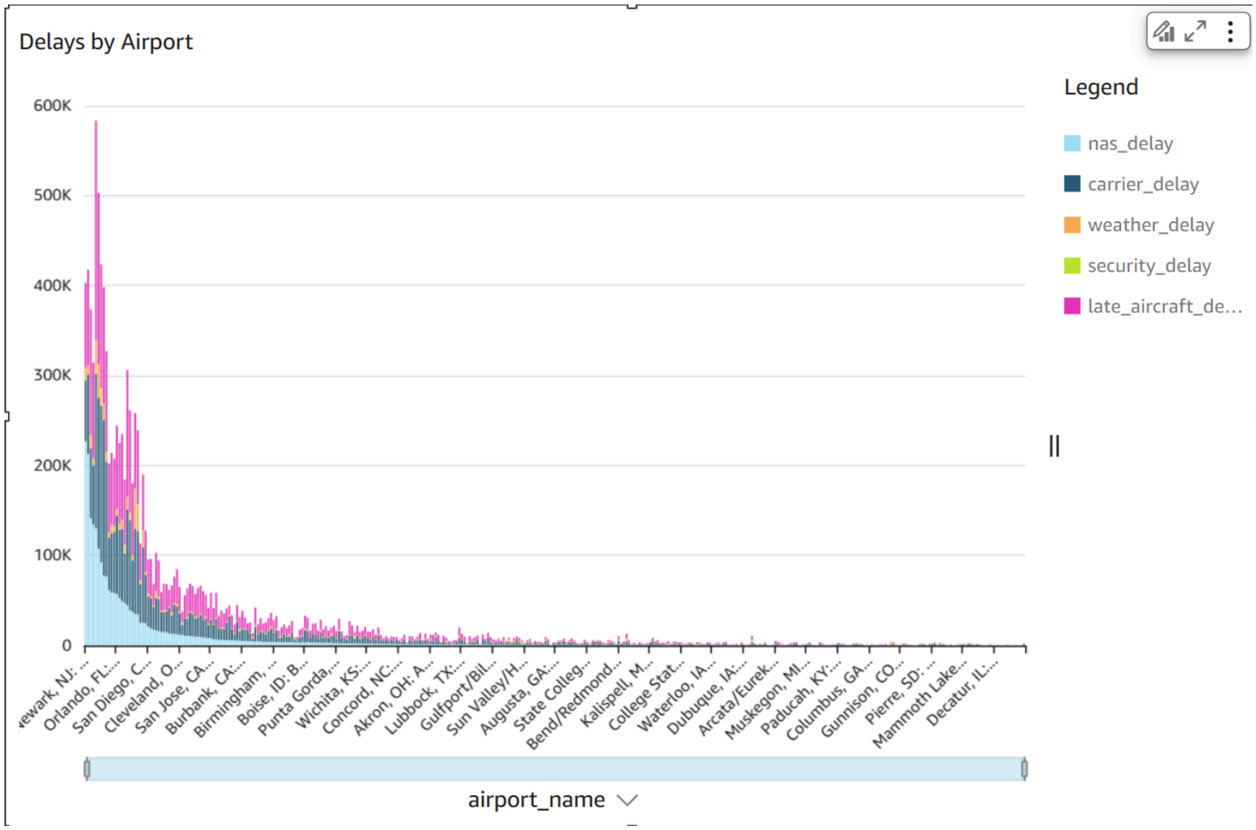
The airports with the highest average arrival delays are San Francisco International (SFO), Houston George Bush Intercontinental (IAH),Boston Logan International followed by Tampa international Oakland international

4. **Delay Causes**: What are the most common causes of delays, and how do they differ by airline and airport?

   Delays by Airline

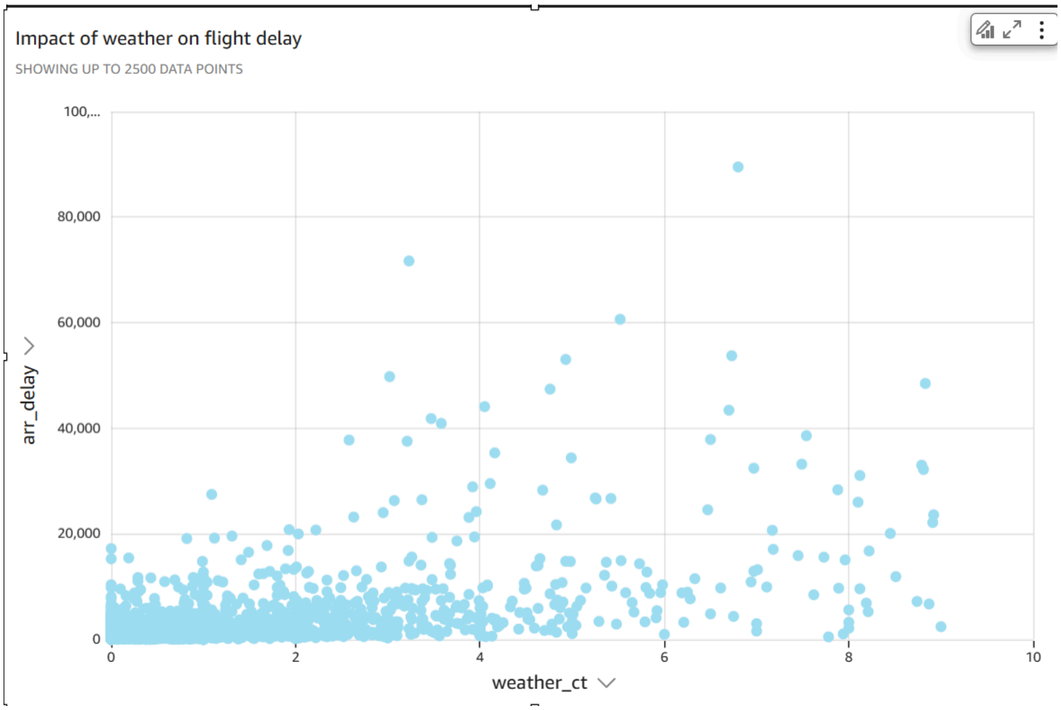


This stacked bar chart visualizes total flight delays by airline, categorized by delay type. JetBlue Airways, SkyWest Airlines, and Southwest Airlines have the highest total delays, with significant contributions from carrier delays (dark blue) and late aircraft delays (pink). Delta Air Lines, United Airlines, and American Airlines also experience substantial delays, though their distribution appears more balanced across different delay types. Hawaiian Airlines and Allegiant Air have the lowest total delays, indicating better on-time performance or lower flight volumes. Carrier-related and late aircraft delays are the most common causes of flight disruptions across airlines.



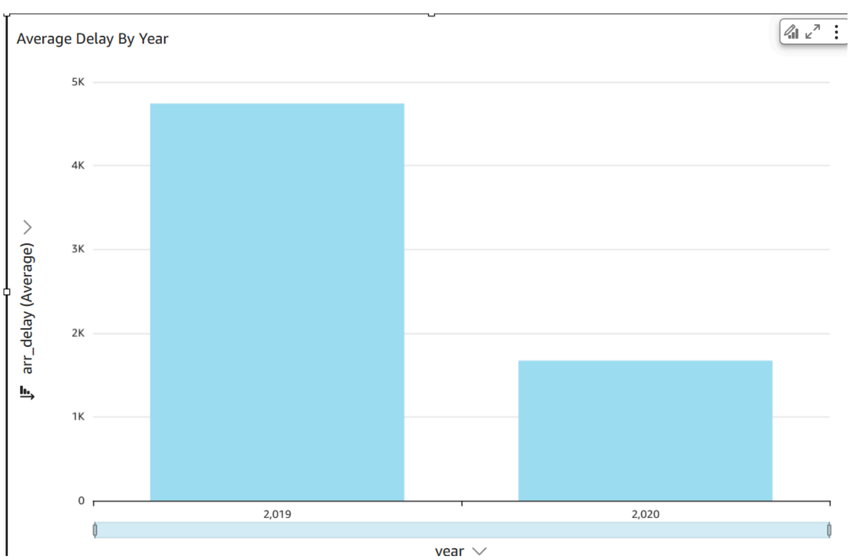
The left side of the chart shows that a few major airports experience the highest total delays, with Newark (EWR), Orlando (MCO), and San Diego (SAN) appearing as some of the most delayed airports. Carrier delays (dark blue) and late aircraft delays (pink) are the dominant causes of delays across most airports. As the graph progresses to the right, smaller airports experience significantly fewer delays. This suggests that larger, high-traffic airports are more prone to delays due to congestion, airline operations, and late arriving aircraft.

5. **Weather Influence**: How does weather impact flight delays, and can it be correlated with delay frequency?



This scatter plot visualizes the impact of weather-related delays (weather_ct) on total arrival delay (arr_delay). The data points show that when weather_ct is low (closer to 0), delays are generally lower, though there are still some outliers with high delays. As weather_ct increases, there is a wider spread of delays, with some flights experiencing extremely long arrival delays exceeding 60,000 minutes. However, there is no strong linear correlation, suggesting that while weather contributes to delays, other factors such as airport congestion and airline operations also play a role. Overall, higher weather-related delays tend to increase variability in arrival delays but do not always guarantee extreme delays.

6. **Forecasting Trends**: Is there a trend in delay duration over time, and can we forecast future delays?



This bar chart compares the average arrival delay between the years 2019 and 2020, showing a significant decrease in delays in 2020. The drop in delays is likely due to the COVID-19 pandemic, which caused a major reduction in air traffic, leading to less congestion at airports and fewer delays overall. In 2019, higher flight volumes and air traffic congestion contributed to longer delays, whereas in 2020, reduced airline operations and improved scheduling may have played a role in minimizing delays.

EXPERIENCE DEVELOPING THE PROJECT AND LESSONS LEARNED

Developing this project came with several challenges, particularly in data availability and querying constraints. One major limitation was that the dataset only contained data from December of 2019 and 2020, making it difficult to analyze trends across different seasons, months, or longer timeframes. Additionally, the dataset lacked destination and arrival airport information, which made it challenging to determine which routes were most prone to delays. Another key issue arose when attempting to query the data from Athena in QuickSight, I encountered a permissions error that prevented me from directly querying the dataset. As a workaround, I uploaded the CSV file into QuickSight and built visualizations from there, which ultimately allowed for better insights, but limited my ability to filter and manipulate data dynamically through queries.

Despite these challenges, some aspects of the project worked particularly well. Creating visualizations was an effective way to analyze and interpret the data, making it easier to identify patterns and outliers, the visualizations in turn made more sense in comparison to the queries. However, if I were to redo the project, I would have transformed the data to create categorized delay groups such as short, medium, and long delays based on time thresholds. This would have made it easier to analyze delay patterns. I would also use sagemaker to create an algorithm that would predict any future delayals if possible.Additionally, I encountered issues with missing values, as the percentage of missing data was initially calculated incorrectly. When I attempted to modify the query, Athena incorrectly reported over 1,000 missing values, which did not align with my expectations. In the future, I would double-check calculations early on and explore alternative SQL approaches to ensure accurate data handling.